Module 7
Predicting Outcomes or Group Membership

Predicted Variable	Number of Predicted Variables	Number of Predictors	Linearity Assumption Met	Possible Techniques
Continuous	1	1	Yes	Simple Linear Regression Analysis
				Robust Techniques (Ridge Regression, Lasso Regression, Elastic Net Regression)
			No	Polynomial Regression
				Quantile Regression
		2+	Yes	Multiple Linear Regression
				Robust Techniques (Ridge Regression, Lasso Regression, Elastic Net Regression)
			No	Polynomial Regression
				Quantile Regression
	2+	1	Yes	Multivariate Linear Regression Analysis
		2+	Yes	Multivariate Linear Regression Analysis
Continuous in a hierarchical or nested structure	1	1	Yes	Multilevel Linear Regression
		2	Yes	Multilevel Linear Regression
Group membership	Categorical (two categories)		Yes	Discriminant Function Analysis
			No	Binary Logistic Regression
				Decision Trees Random Forest Support Vector Machines
	Categorical (multiple categories)		Yes	Discriminant Function Analysis
			No	Multinomial Logistic Regression
				Decision Trees Random Forest Support Vector Machines
	Categorical (ordinal)		Yes	Discriminant Function Analysis
			No	Decision Trees Random Forest Support Vector Machines
	Multiple categorical		No	Multivariate Logistic Regression Multivariate Probit Models Multivariate Random Forest
Group membership in a hierarchical or nested structure	Categorical (two categories)		No	Multilevel Logistic Regression
	Categorical (multiple categories)		No	Multilevel Logistic Regression
	Categorical (ordinal)		No	Multilevel Ordinal Logistic Regression
	Multiple categorical			